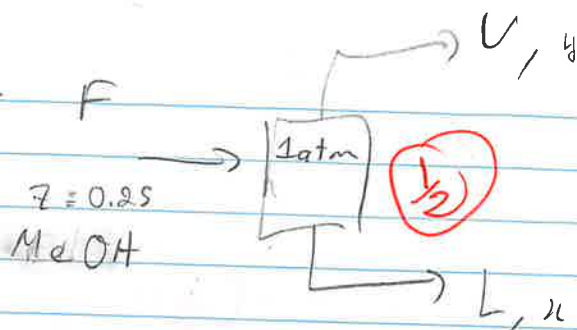


Quinn Hays, L
CHE 313 HW I

①

3

Q. 1:



Balance eqn:
$$\begin{cases} F = V + y \\ Fz = Vy + Lx \end{cases}$$

$$y = -\frac{L}{V}x + \frac{F}{V}z$$

a) $f = 1 \Rightarrow \frac{V}{F} = 1 \Rightarrow -\frac{L}{V} = \frac{-(F-V)}{V} = 0$

$\Rightarrow y = 0x + z \Rightarrow y = z$

From y - x curve, $y = 25\%$, $x = 5\%$

$\Rightarrow y = 0.25, x = 0.05$

b) $f = 0 \Rightarrow \frac{V}{F} = 0 \Rightarrow -\frac{L}{V} = \frac{-(F-V)}{V} = \frac{-F}{0} = -\infty$

$\Rightarrow$ vertical line at $x = 0.25$

From y - x curve, $y = 63\%$, $x = 25\%$

$y = 0.63, x = 0.25$

c) $f = 1/3 \Rightarrow \frac{V}{F} = \frac{1}{3} \Rightarrow \frac{L}{V} = \frac{2}{3}$

$\Rightarrow y = -\frac{2}{3}x + 3 \times 0.25 \Rightarrow y = -\frac{2}{3}x + 0.75$

From y - x curve, $y = 0.49, x = 0.14$

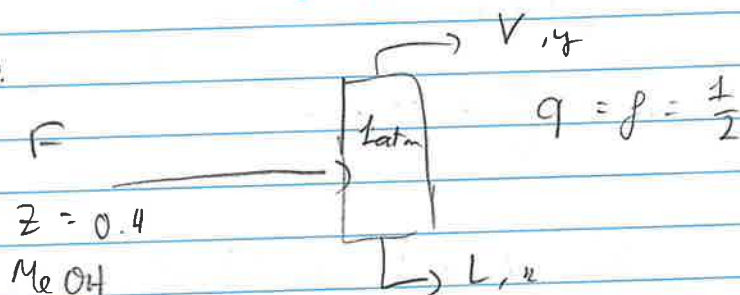
$$d) f = \frac{2}{3} \Rightarrow \frac{V}{F} = \frac{2}{3} \Rightarrow \frac{L}{F} = \frac{1}{3} \Rightarrow \frac{-L}{V} = -\frac{1}{2}$$

$$\Rightarrow y = -\frac{1}{2}x + \frac{3}{2} \times 0.25 \Rightarrow y = -\frac{1}{2}x + 0.375$$

From graph, $y = 0.35, x = 0.07$ 1/2

graph 1/2

Q.2:



Balance eqn:
$$\begin{cases} F = L + V \\ Fz = Lx + Vy \end{cases}$$

$$y = \frac{-L}{V}x + \frac{F}{V}z$$

$$a) f = \frac{1}{2} \Rightarrow \frac{V}{F} = \frac{1}{2} \Rightarrow \frac{L}{F} = \frac{1}{2} \Rightarrow \frac{-L}{V} = -1$$

$$\Rightarrow y = -x + 2 \times 0.4$$

From graph, $y = 0.59, x = 0.21$ 9

$\Rightarrow$ Vapour stream contains 59 mol% MeOH, liquid stream contains 21 mol% MeOH

b) + c) Check extreme cases ($f = 1$ and $f = 0$)

$$f = 1 \Rightarrow \frac{V}{F} = 1 \Rightarrow \frac{L}{F} = 0 \Rightarrow \frac{-L}{V} = 0 \Rightarrow y = z = 0.4$$

drawing this line on graph gives $y = 0.4, x = 0.09$

$$f = 0 \Rightarrow \frac{V}{F} = 0 \Rightarrow \frac{L}{F} = 1 \Rightarrow \frac{L}{V} = \infty \Rightarrow \text{vertical line at } x = 0.4$$

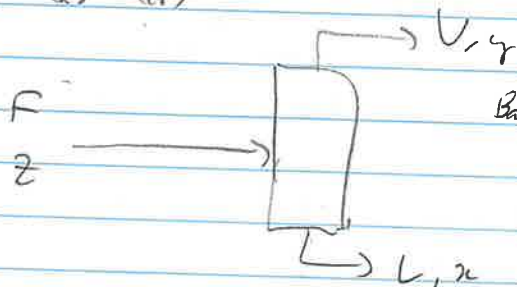
This yield $x=0.4$, $y=0.729$.

$\Rightarrow x_{\min} = 0.09$, $x_{\max} = 0.4$

$L_{x_{H_2O \max}} = 0.91$

max MeOH in liquid stream is 40 mol %
max H_2O in liquid stream is 91 mol %

Q.3: 2.D.1 (a) $\rightarrow$ (d)



Balance eqn: $F = V + L$

$Fz = Vy + Lx$

$y = -\frac{L}{V}x + \frac{F}{V}z$

a) $F = 100.0 \text{ kmol/h}$

$z = 0.600$

$f = \frac{V}{F} = 0.400 \Rightarrow \frac{L}{F} = 0.600 \Rightarrow -\frac{L}{V} = -\frac{0.600}{0.400} = -\frac{3}{2}$

$\Rightarrow y = -\frac{3}{2}x + \frac{1}{0.400} \times 0.600$

From graph, $y=0.760$, $x=0.490$

$\Rightarrow 100.0 = V + L$

$0.600 \times 100 = 0.760V + 0.490L$

Solve system above yield:

$V = 40.7 \text{ kmol/h}$
 $L = 59.3 \text{ kmol/h}$

b) For $F = 1500.0 \text{ kmol/h}$

We still have $y=0.760$, $x=0.490$

$\Rightarrow 1500 = V + L$

$0.600 \times 1500 = 0.760V + 0.490L$

$V = 611.1 \text{ kmol/h}$
 $L = 888.9 \text{ kmol/h}$

c) $z = 0.300$

$x = 0.200$

$F = 1000.0 \text{ lbmol/h}$

From y - x curve, $x = 0.200$ correspond to $y = 0.579$

$\Rightarrow$ Solve balance eqns: $\begin{cases} 1000 = V + L \\ 1000 \cdot 0.200 = 0.579V + 0.200L \end{cases}$

$\Rightarrow \boxed{\begin{matrix} V = 263.9 \text{ lbmol/h} \\ L = 736.1 \text{ lbmol/h} \end{matrix}}$

$\Rightarrow \frac{V}{F} = \frac{263.9}{1000.0} \Rightarrow \boxed{\frac{V}{F} = 0.2639}$

$y = -\frac{L}{V}x + \frac{F}{V}z$

Alternatively:

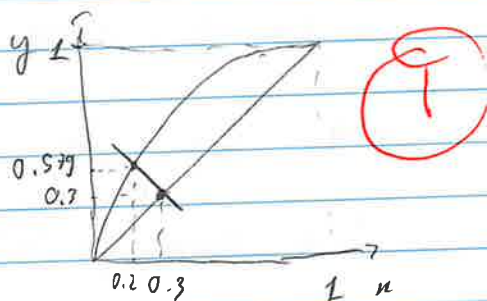
line $y = -\frac{L}{V}x + \frac{F}{V}z$

$\Rightarrow \frac{-L}{V} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0.3 - 0.579}{0.3 - 0.2}$

$\Rightarrow \frac{L}{V} = 2.79$

$\Rightarrow \frac{L/F}{V/F} = 2.79 \Rightarrow \frac{L}{F} = 2.79 \frac{V}{F} \Rightarrow \frac{L}{F} + \frac{V}{F} = 1 \Rightarrow 3.79 \frac{V}{F} = 1$

$\Rightarrow \boxed{\frac{V}{F} = 0.2639} \rightarrow \text{identical as above}$



d) $x = 0.45$

$L = 1500 \text{ kmol/h}$

$V/F = 0.2 \rightarrow V = 0.2F = 300 \text{ kmol/h}$

From curve: $x = 0.45 \Rightarrow y = 0.75$

$\Rightarrow$ Balance eqns: $\begin{cases} F = V + L \\ Fz = Vy + Lx \end{cases} \Rightarrow \begin{cases} F = 0.2F + 1500 \\ Fz = 0.2F \cdot 0.75 + 1500 \cdot 0.45 \end{cases} \Rightarrow \boxed{\begin{matrix} F = 1875 \text{ kmol/h} \\ z = 0.51 \end{matrix}}$

Table 2-8, Q.1

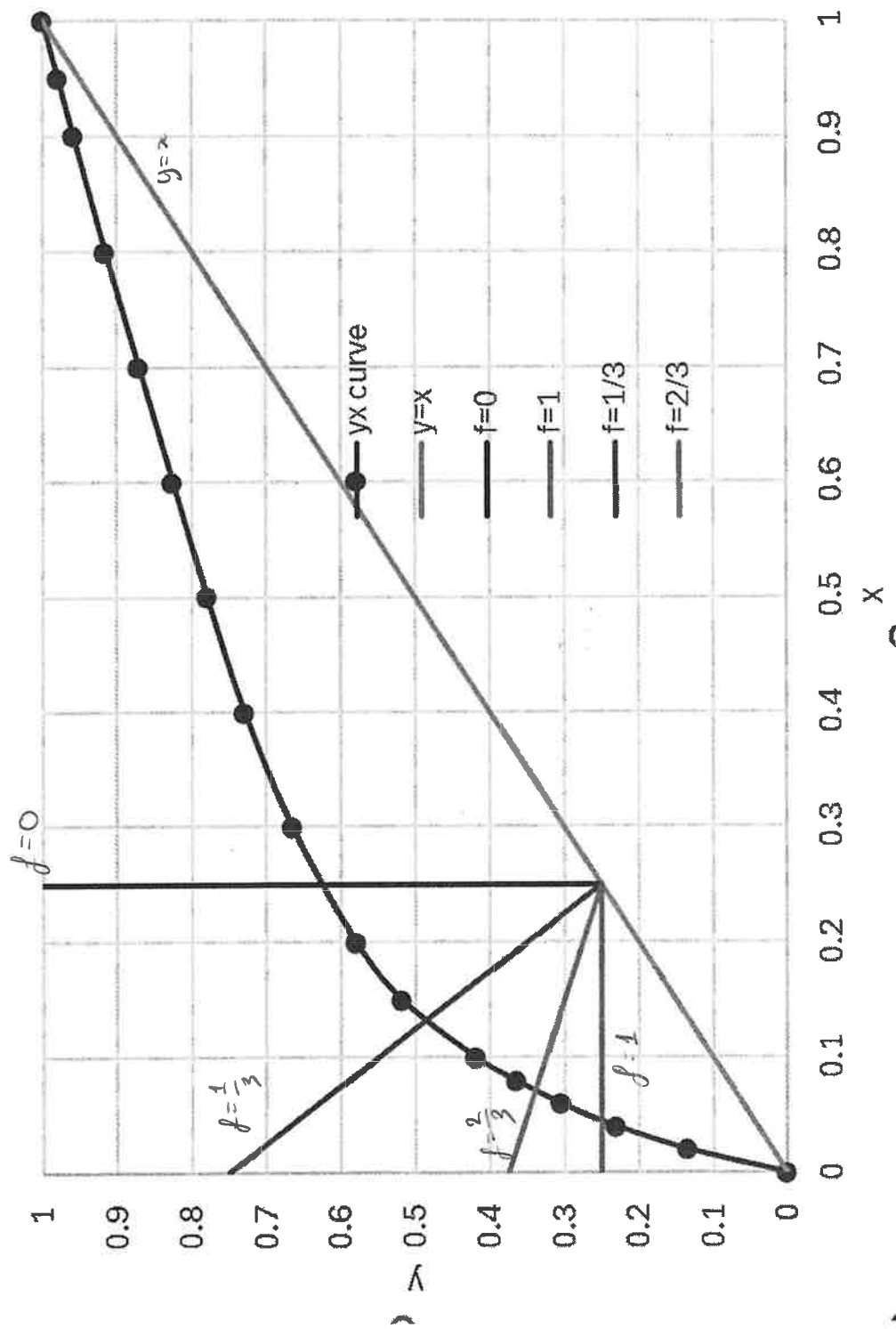


table 2-8 Q.2

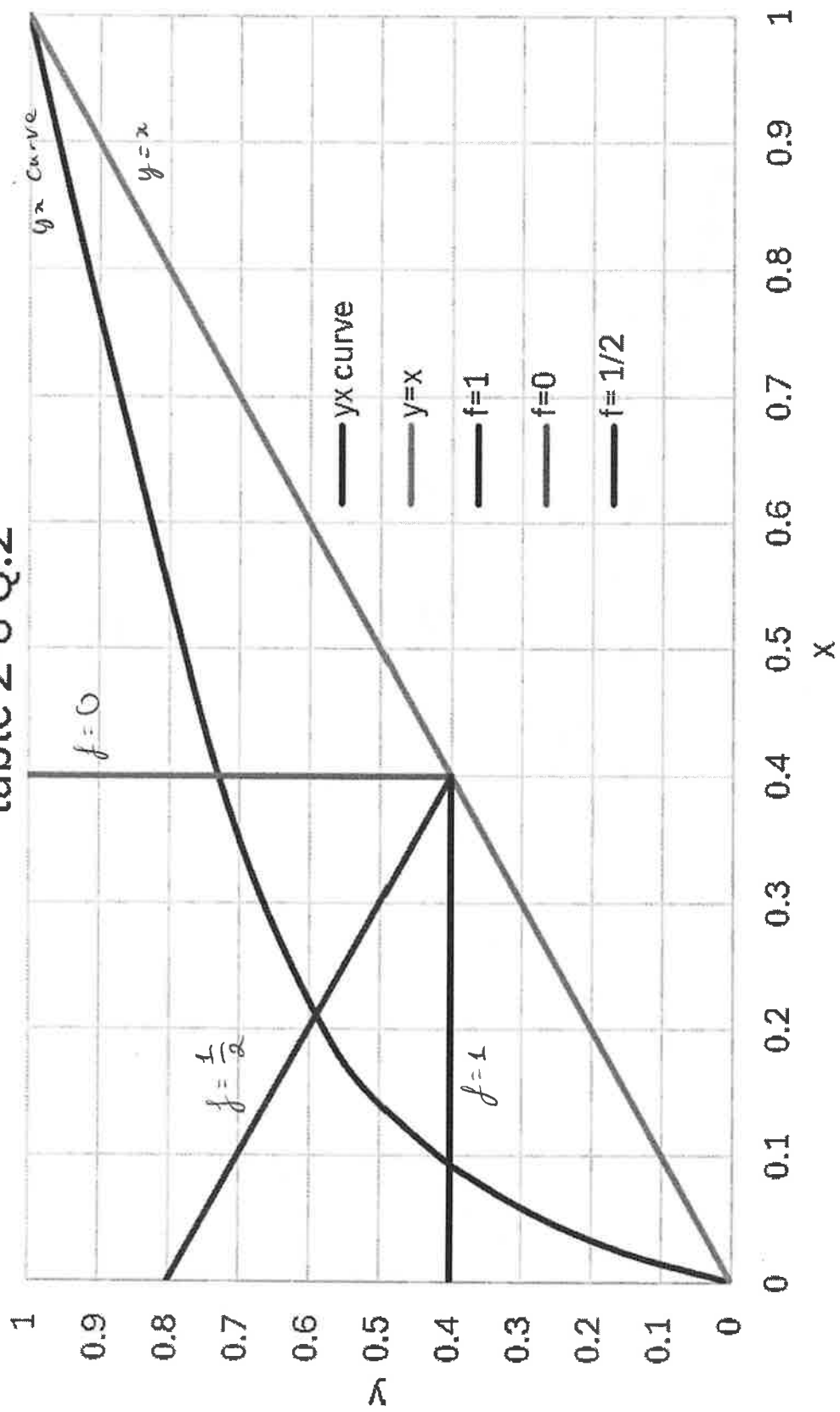


table 2-8 Q.3

